Arjun Garg

Senior Full Stack & AI/ML Engineer

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PROFILE

- Senior Full Stack Engineer and AI/ML Engineer with 10+ years of experience delivering scalable, cloud-native, and AI-enhanced platforms across e-commerce, energy, and healthcare.
- Strong focus on backend systems (Java 17, Kotlin, Python, Spring Boot, FastAPI), MLOps (MLflow, SageMaker, Airflow), and real-time data architectures (Kafka, Redis Streams).
- Experienced integrating LLMs, ranking models, and search pipelines with LangChain, PyTorch, and FAISS. Proficient in deploying microservices to AWS, Azure, and GCP using Kubernetes, Terraform, and CI/CD pipelines.
- Recognized for bridging ML engineering and production systems to drive real-world impact — reducing latency by 45%, boosting personalization accuracy by 30%, and serving millions of users securely and reliably.

EDUCATION

Bachelor's Degree in Computer Science 2010 – 2014
University of Illinois at Urbana-Champaign

CERTIFICATES

Azure Databricks Platform Architect: Issued May 2024 - Expires May 2026 | Credential ID: 104090644

PROFESSIONAL EXPERIENCE

Senior Software Engineer (AI/ML focus) / Tech Lead 07/2020 – Present | OR, US
Nike

- Led development of Java 17 and Kotlin microservices using Spring Boot 3.x, deployed across AWS EKS and Lambda for global e-commerce flows.
- Integrated Python 3.11 services using Django and FastAPI for loyalty logic, real-time personalization, and recommendation engines.
- Built Go 1.21-based Gin APIs for high-frequency product availability requests; reduced average response time from 640ms to under 260ms.
- Designed and exposed REST and GraphQL APIs consumed by React 18 and Vue.js 3 frontends; frontend latency dropped by 35%.
- Served PyTorch 2.x models using FastAPI endpoints and TorchServe, cutting inference cold start time by 41%.

- Delivered a LangChain + FAISS-based Q&A system that resolved 30% more order-related tickets automatically within 2 weeks.
- Built event-driven Kafka 3.x pipelines for loyalty events and abandoned cart triggers, improving throughput by 37%.
- Implemented aggressive caching and concurrency tuning across API layer; boosted ML response success rate from 89.2% → **96.7%**.
- Reduced p95 checkout API latency from 740ms → **320ms** and improved order placement reliability to 99.98%.
- Scaled LangChain-powered LLM assistant to 400K+ users with sustained >98% system-wide response accuracy.
- Automated all CI/CD pipelines using GitHub Actions and Terraform; cut deployment time from 20 min → **7 min** with zero-downtime rollouts.
- ✅ **Improvement Summary:** Boosted backend response rate from 91.5% → **98.2%**, reduced ML latency by **41%**, and increased personalization conversion by **22.6%**.

Senior Java Full Stack Engineer / ML Platform Developer 01/2017 – 06/2020 | Chicago, IL

Exelon

- Migrated legacy Java EE monoliths to modular Spring Boot microservices running on Azure AKS and GCP GKE.
- Developed scalable Java 11 and Kotlin services to support smart meter data ingestion and load analytics.
- Built Python FastAPI endpoints exposing TensorFlow and XGBoost models for peak demand forecasting.
- Integrated Kafka 2.x pipelines and Redis Streams for ingesting ERCOT and PJM live market data.
- Created Angular-based internal dashboards with real-time visualizations of voltage, load, and outage conditions.
- Built Android API interfaces for technician tools that enabled on-site grid diagnostics and fault logging.
- Used Apache Airflow to automate daily ETL processes for training set generation from Redshift and BigQuery.
- Designed and enforced secure authentication using OAuth2.0, JWT, and Spring Security with LDAP RBAC.
- Instrumented Prometheus + Grafana for real-time alerting on ML drift, data staleness, and API downtime.
- Led technical workshops on integrating ML pipelines into grid systems for internal data teams.
- Tuned Flink windowing logic to reduce event duplication in power usage aggregation jobs.
- ✅ **Improvement Summary:** Reduced load forecasting error rate from **12.7% → 6.3%**, leading to better grid balancing and fewer false outage escalations.

Full Stack Developer 01/2015 – 12/2016 | MO, US

Optum

Projects: OptumCare Patient Management, Healthcare Analytics

- Designed Java 8 backend services using Spring Boot for patient claims, provider access, and FHIR endpoints.
- Created real-time patient risk dashboards using AngularJS and Angular integrated with ML predictions.
- Built Python FastAPI APIs serving XGBoost-based risk scoring models for chronic disease prediction.
- Streamed Kafka clinical event data and processed incoming HL7 lab feeds using Quartz Scheduler and Redis.
- Exposed clinical scoring services securely via OAuth2.0, Spring Security, and JWT-based authentication.

- Deployed HIPAA-compliant microservices to AWS ECS and Fargate; integrated CloudWatch for logging.
- Automated CI/CD pipelines with Jenkins and GitHub Actions; reduced rollout effort and deployment errors.
- ✅ **Improvement Summary:** Improved chronic disease prediction precision from **81.4% → 90.2%**, significantly enhancing provider trust in model outputs.

SKILLS

Languages
Java (8–17), Python (3.8–3.11), Go (1.21), Kotlin, Scala, TypeScript, JavaScript (ES6+), Bash, SQL

Frontend Development
React (18), Angular (6–14), Vue.js (3), AngularJS (1.7), TypeScript, Tailwind CSS, Bootstrap, Material UI

Data Engineering & Orchestration
Apache Kafka (2.x–3.x), Redis Streams, RabbitMQ, Apache Flink, Apache Airflow (2.x), Apache Spark,PostgreSQL, MySQL, MongoDB, Cosmos DB, Redshift, Snowflake, BigQuery, DataLake (S3 / ADLS)

Monitoring & Observability
Prometheus (2.x), Grafana (10), ELK Stack (Elasticsearch 8.x, Logstash, Kibana), Azure Monitor,Datadog, CloudWatch Logs, MLflow for model tracking & drift

Streaming & Real-Time Pipelines
Apache Kafka (2.x–3.x), Kafka Connect, Kafka Streams, RabbitMQ, Redis Streams, Apache Flink, Apache Airflow (2.x), Apache Spark

AI/ML & Generative AI
PyTorch (2.x), TensorFlow (2.x), XGBoost, Scikit-learn, MLflow (2.x), LangChain (v0.1+), FAISS,RAG Pipelines, FastAPI (0.75–0.103), Django (4.x), TorchServe, Hugging Face Transformers,Prompt Engineering, OpenAI APIs (GPT-3.5 / GPT-4), CrewAI, Autogen, LLM fine-tuning, LLMOps,Model Drift Detection, Retrieval-Augmented Generation (RAG), LLM Evaluation Pipelines

Mobile Development
React Native, Android (Kotlin), iOS (Swift), FHIR app integrations, Barcode scanning apps,Offline-first mobile sync, Custom SDK integrations, Hybrid webview APIs

Databases
PostgreSQL, MySQL, MongoDB, Cosmos DB, Redis, Oracle DB, Snowflake, Redshift, BigQuery, Azure SQL, DynamoDB

MLOps & CI/CD
MLflow, SageMaker Pipelines, TorchServe, TensorFlow Serving, Airflow ML retraining,Dockerized model builds, Versioned model registry, Canary deployments, Zero-downtime rollouts, GitOps

Backend & API Development
Spring Boot (1.5–3.2), FastAPI (0.75+), Django (4.x), Flask, Node.js, Express.js, Gin (Go 1.21),RESTful APIs, GraphQL, gRPC, HL7, FHIR, JSON/REST spec design, OAuth-secured APIs

Cloud & DevOps
AWS (EKS, ECS, Lambda, SageMaker, S3, CloudWatch, API Gateway), Azure (AKS, Blob Storage, Functions, DevOps Pipelines),GCP (BigQuery, GKE, Vertex AI), Docker, Kubernetes, Helm, Terraform (1.6+), Jenkins, GitHub Actions, GitLab CI, JFrog Artifactory

Security & Compliance
Spring Security, OAuth 2.0, OpenID Connect, JWT, LDAP, RBAC, SSO, API Key rotation, HIPAA, PCI-DSS, Data Encryption (AES256/SSL)

Tooling & Collaboration
Git, GitHub, GitLab, Bitbucket, JIRA, Confluence, Figma, Slack, Agile/Scrum

📂 PROJECTS

Nike

1. **Digital Checkout and Loyalty Microservices**
   - Developed and led Kotlin/Java-based microservices on AWS EKS for checkout, loyalty events, and cart management.
   - Reduced p95 latency from 740ms → 320ms and improved reliability to 99.98%.
   - Integrated Kafka pipelines for cart abandonment and loyalty tiers.
2. **LLM-Powered Order Help Assistant**
   - Created a LangChain + FAISS RAG pipeline behind FastAPI for automated post-order Q&A and product support.
   - Served PyTorch models via TorchServe; increased resolution accuracy by 30%.
   - Scaled usage to 400K+ users with <2s average latency and 98.2% backend response rate.
3. **Real-Time Personalization Engine**
   - Collaborated with ML team to build Django + FastAPI APIs for real-time ranking and reward personalization.
   - Model cold-start time improved by 41%; personalization conversion rose by 22.6%.

Tech Stack: Java 17, Kotlin, Spring Boot 3.x, Python 3.11, Django 4.x, FastAPI 0.103, Go 1.21, React 18, Vue.js 3, Kafka 3.x, AWS EKS, MLflow, Terraform

Exelon

1. **Smart Grid Load Forecasting**
   - Built Java + Python services to stream ERCOT/PJM data into ML pipelines and serve real-time predictions.
   - Deployed FastAPI scoring endpoints for TensorFlow models; forecast error dropped from 12.7% → 6.3%.
2. **Technician Tools + Live Grid Analytics**
   - Developed Angular dashboards and Android APIs used in the field for diagnostics and repair events.
   - Integrated real-time alerts from Kafka and Flink; added RBAC-secured access for mobile ops staff.
3. **Model Monitoring + Data Drift Detection**
   - Used Airflow to retrain models weekly on Redshift + BigQuery; implemented MLflow tracking and Grafana dashboards.
   - Improved data pipeline turnaround from 24h → 3h; supported ML SLA reliability >96%.

Tech Stack: Java 11, Spring Boot 2.x, Python, FastAPI, TensorFlow, XGBoost, Kafka 2.x, Angular, Airflow, Azure AKS, MLflow, Redis Streams

Optum

1. **Risk Scoring & Clinical Alerts Platform**
   - Built Spring Boot services and FastAPI ML APIs to deliver real-time chronic disease predictions.
   - Model precision improved from 81.4% → 90.2%; adopted by 5+ internal care teams.
2. **Kafka-Driven Event Processing & ETL**
   - Streamed HL7 + lab feeds via Kafka and Quartz; alerts surfaced in Angular dashboards.
   - Automated retraining workflows with Airflow and reduced latency of clinical event alerts from 3h → 30m.

Tech Stack: Java 8, Spring Boot, Python, FastAPI, XGBoost, AngularJS, Angular, Kafka, Jenkins, AWS Fargate